Code Coverage Report

Project: Online Quiz System
Environment: MSYS2 UCRT64, GCC 16.2.0
Test result: 9/9 test programs passed

Measured Scope

Coverage was measured across all production C++ files under src/ after a clean rebuild and complete test-suite run.

```
gcovr --root . --filter "src/.*\.cpp$" --print-summary
```

Metric	Covered	Total	Percentage
Lines	1,151	1,318	**87.3%**
Functions	58	61	**95.1%**
Raw branches	1,413	2,499	**56.5%**
Normalized branches	1,300	1,573	**82.6%**

Normalized branch coverage excludes compiler-generated exception and unreachable branches:

```
gcovr --root . --filter "src/.*\.cpp$" \
  --exclude-throw-branches --exclude-unreachable-branches --print-summary
```

Coverage Status

The verified full-project result is **87.3% line coverage**, **95.1% function coverage**, **56.5% raw branch coverage**, and **82.6% normalized branch coverage**. Remaining gaps are mainly server startup/concurrency failures, platform-specific paths, and forced filesystem failures.

Branch Coverage Report

Project: Online Quiz System
Environment: MSYS2 UCRT64, GCC 16.2.0
Scope: All production .cpp files under src/

Test Evidence

The complete suite passed **9/9 test programs**, covering quiz parsing, scoring, registration, authentication, security logging, administrator workflows, interactive menu paths, TCP submission, and server socket handling.

Measured Result

Branch metric	Covered	Total	Percentage
Raw branch coverage	1,413	2,499	**56.5%**
Excluding compiler-generated/unreachable branches	1,300	1,573	**82.6%**

The raw value is the unfiltered GCC/gcovr result. The normalized value excludes compiler-generated exception and unreachable branches that are not meaningful application decisions.

Remaining Gaps

- Server startup and concurrency-limit failures
- Platform-specific Windows/POSIX branches
- Forced filesystem and socket failure paths
- Some interactive menu error branches

Both raw and normalized values are reported; the raw full-project branch result is **56.5%**.